

BUYERS' GUIDE

ESTYLE

WIRELESS HEADPHONES Cut The Cord



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We have already deep-dived into the age of 'compact' and 'hands-free,' where "less is the new more." Even headphones manufacturers have found a way to brilliantly improve their products to fit the desires of customers.

Apple's bold step to exclude the conventional 3.5mm headphone jack from its latest iPhone 7 smartphone series for the purpose of space utility maximisation and water-resistance is a case in point. New players too are not far behind. For instance, Seattle-based Human Inc. is coming up with a new product called 'Sound' that gestates touch-based earphone control along with language translation and multiple audience connect. Such developments give reasons enough to believe that the future of wireless headphones is now.

How Bluetooth headphones work

Wireless headphones or earphones work on digitally transmitted waves—via infrared (IR), radio-frequency (RF) waves or Bluetooth—from source. While each device has its own benefits and drawbacks, in this article we will restrict ourselves to Bluetooth-based headphones as these are portable, flexible and in demand.

Bluetooth-driven wireless headphones consist of a receiver that receives the signal transmitted from the Bluetooth device, a small amplifier to strengthen the signal and a battery pack to run the system. The device with Bluetooth (smartphone, tablet, laptop among others) has a small computer chip inside that contains the Bluetooth radio module and software that enable the two devices to pair. Bluetooth digitally transmits the audio through a low-power radio signal, which is received by the receiver of the headphone.

The checklist

Sound quality. Composed of magnet, voice coils and a diaphragm, the driver unit of a headphone converts electrical energy into sound pressure. Larger-diameter drivers mean

bigger speakers, consequently giving stronger audio output. However, the output quality directly depends on driver components. On an average, driver diameter is 8 to 12mm for earphones and 20 to 50mm for headphones.

Sound quality can be judged from factors like sensitivity and sound pressure level (SPL), impedance and total harmonic distortion (THD). SPL decides the loudness range of a headphone. On an average, most headphones have an SPL in the range of 85 to 120dB SPL/mW. Scale of impedance (the resistance to the current that creates the sound) decides the life of the device. Impedance of basic-level headphones tends to be lower (around 32 ohms), requiring less current to work. But this may create background noise. High-end headphones have a higher impedance (up to 330 ohms) and require internal amplifiers for operation, but this also prevents early headphone damage. THD level indicates distortion of sound under high volume. Lower THD is always desirable, and on an average, headphones have less than 1% THD.

Design. Select the headphone design as per your intended application, comfort and personal preferences. Headphones come in in-ear, on-ear and over-ear designs. In-ear (canal) design sits in the ear canal directly, providing more impactful sound and better exclusion of external noise. In on-ear (supra-aural) design, pieces rest on top of the ear, preventing trapping of heat on the ears. This design is, however, less effective in noise blocking. Over-ear (circumaural) design covers ears completely, ensuring better quality and range of sound. The huge size of the design accommodates a bigger engine, amplifying the sound output (especially the bass experience) by multi-fold.

Battery life. Wireless headphones may come with a built-in rechargeable or disposable battery, the former being more common in popular brands. On an average, they use



Wireless headphone designs—in-ear, on-ear and over-ear (from top to bottom)